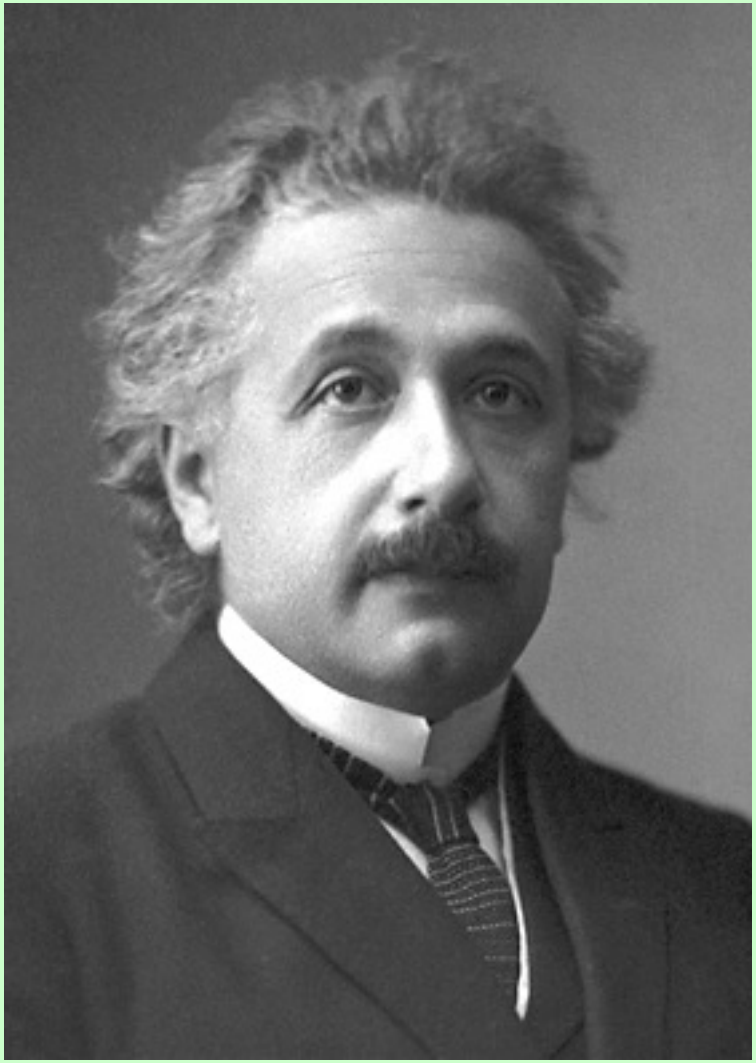
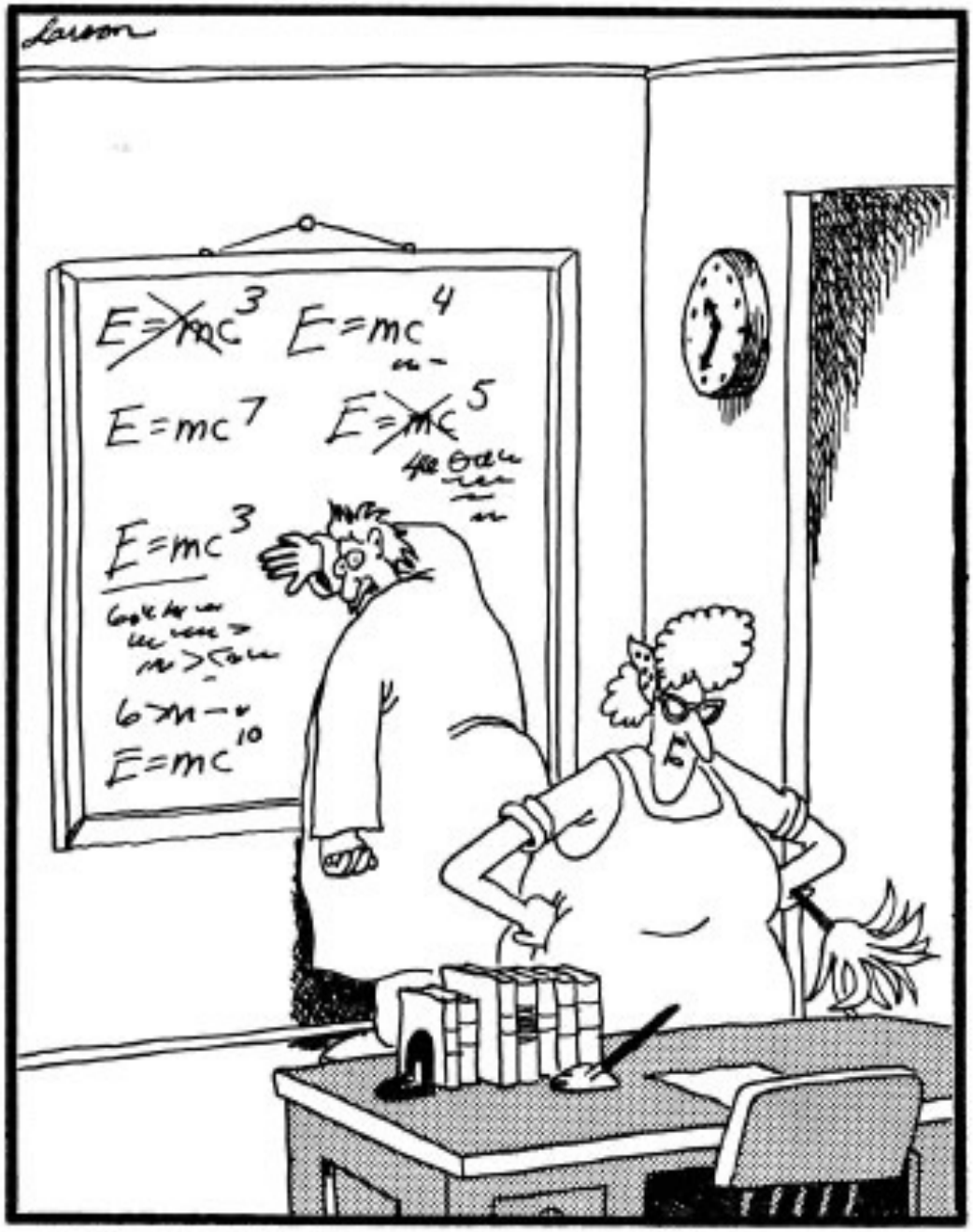


Special Relativity

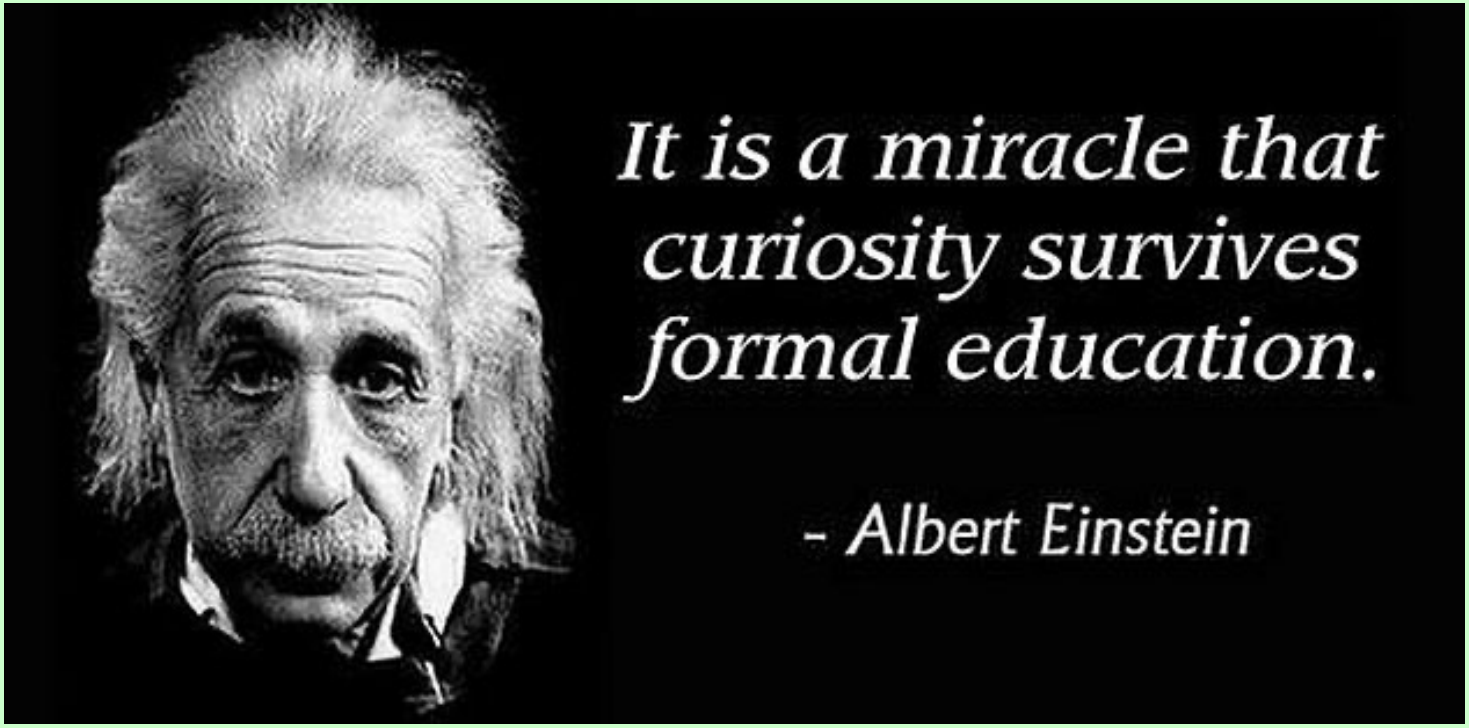


"*Now* that desk looks better. Everything's squared away, yessir, squaaaaaared away."

Ex. What is the principle of Galilean Relativity? What about light seemed to challenge it? What did Einstein Propose as the solution?

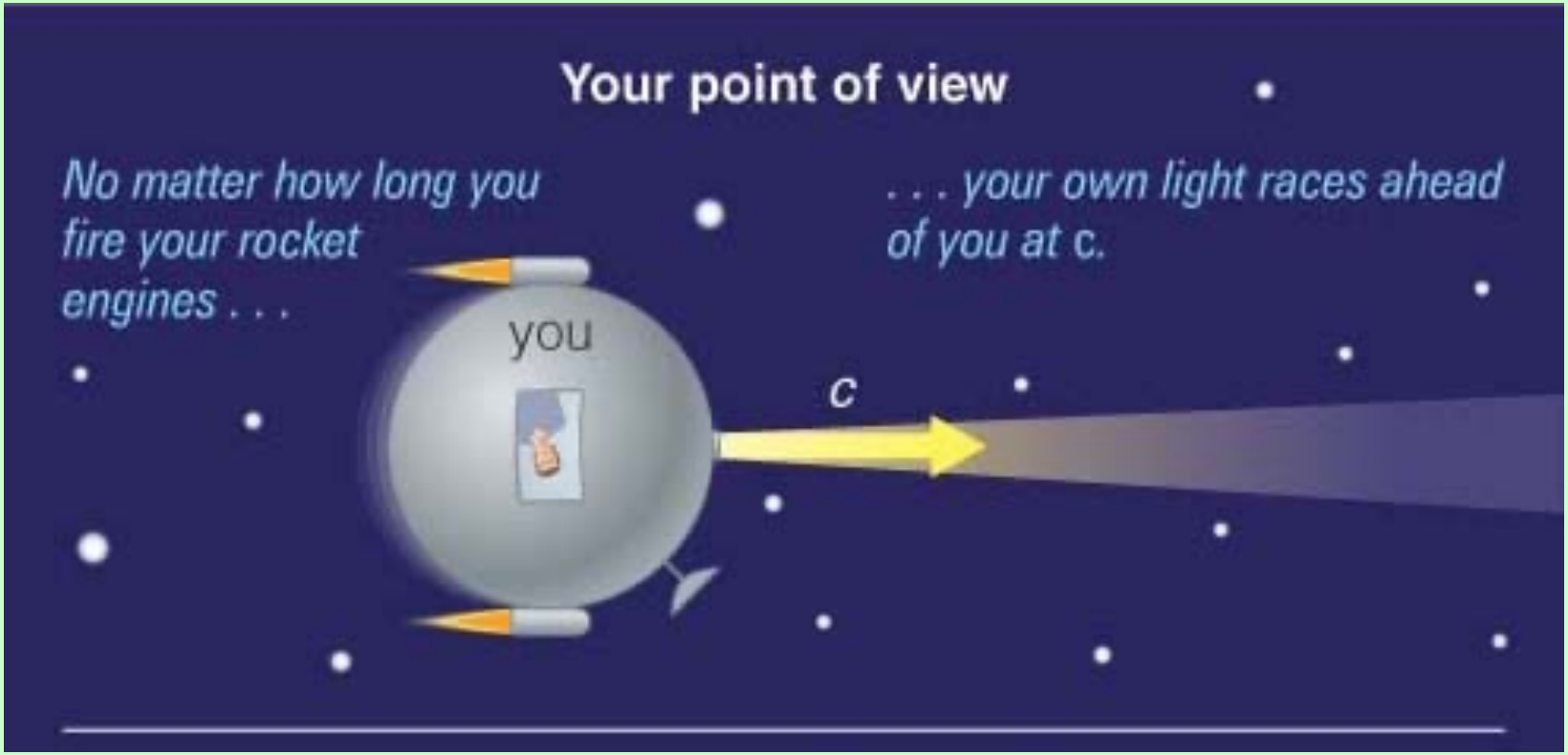
Postulates of Special Relativity

1. The laws of nature are the same in all inertial reference frames
2. The speed of light in a vacuum is the same in all inertial reference frames



What's special about SR? Its when all reference frames are not accelerating and are IRFs
General Relativity deals with the case of accelerating Frames

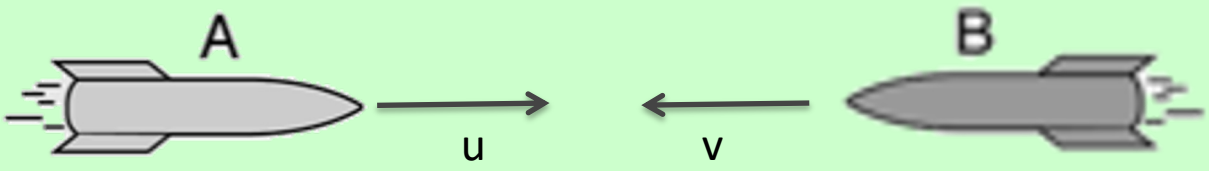
Consequence #1: Velocity Addition



Group Questions:

1) A laser pulse travels for 1 second, how far does it go?
2) What if you are running after it at 50% the speed of light?

More generally, velocities add weirdly in relativity



Adding two velocities:

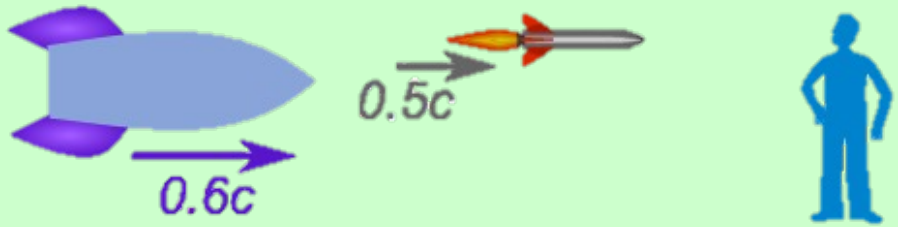
$$\boxed{u' = \frac{u - v}{1 - \beta_u \beta_v}}$$

Make the top of the equation what the answer "should" be

Relativistic beta: $\beta = \frac{v}{c}$

So, for example if v = 2 x 10^8 m/s, then v = 2/3 c, but β= 2/3

Ex. A spaceship moving toward the person at 0.6c launches a missile at 0.5c in the reference frame of the ship. How fast does it approach the person?



A. **Intuition Way:** I need to make the top be the answer I "expect"

$$\boxed{u' = \frac{u - v}{1 - \beta_u \beta_v}}$$

$\boxed{u - v = +1.1c}$

How can I make u-v = 1.1c?

$$u = 0.6c \text{ and } v = -0.5c$$

Plug u and v values in:

$$u' = \frac{0.6c - (-0.5c)}{1 - (.6)(-0.5)} = \frac{1.1c}{1 + .6 * .5} = \frac{1.1c}{1.3} = \boxed{0.846c}$$

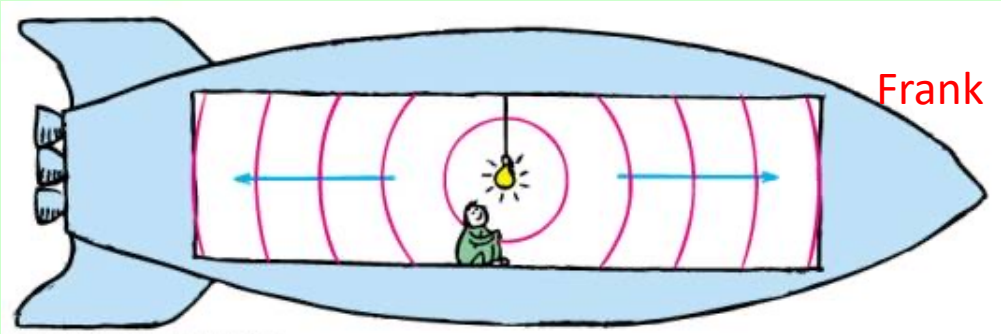
B. **Explaining the same thing with formal reference frames**. What is S and S' ?

S	S'
Ship	MAN
Missile	$V_{man,ship} = -0.6c$
$u = +0.5c$	
$\beta_u = 0.5$	$\beta_v = -0.6$

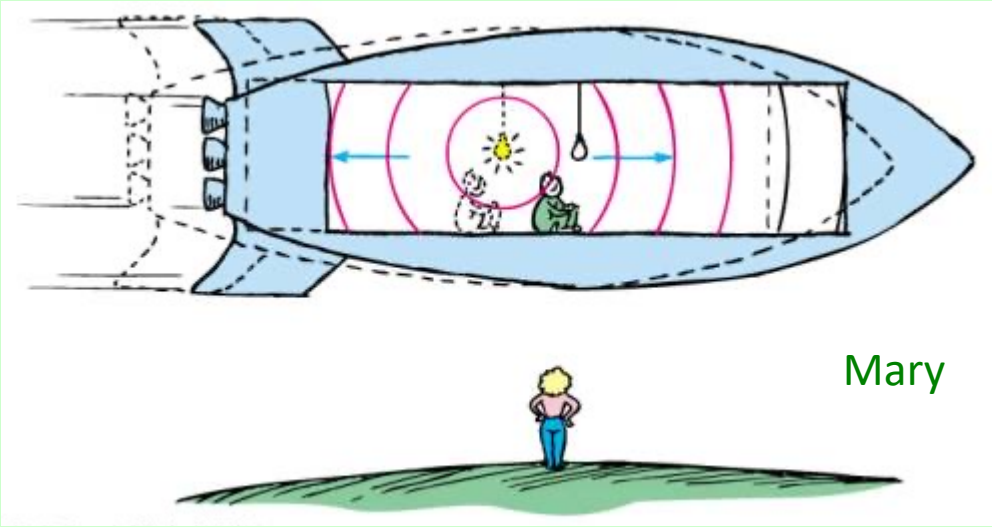
$$u' = \frac{u - v}{1 - \beta_u \beta_v} = \frac{0.5c - (-0.6c)}{1 - (0.5)(-0.6)} = \boxed{0.846c}$$

Consequence #2: Simultaneity

Does light hit the back first, or both ends at together?

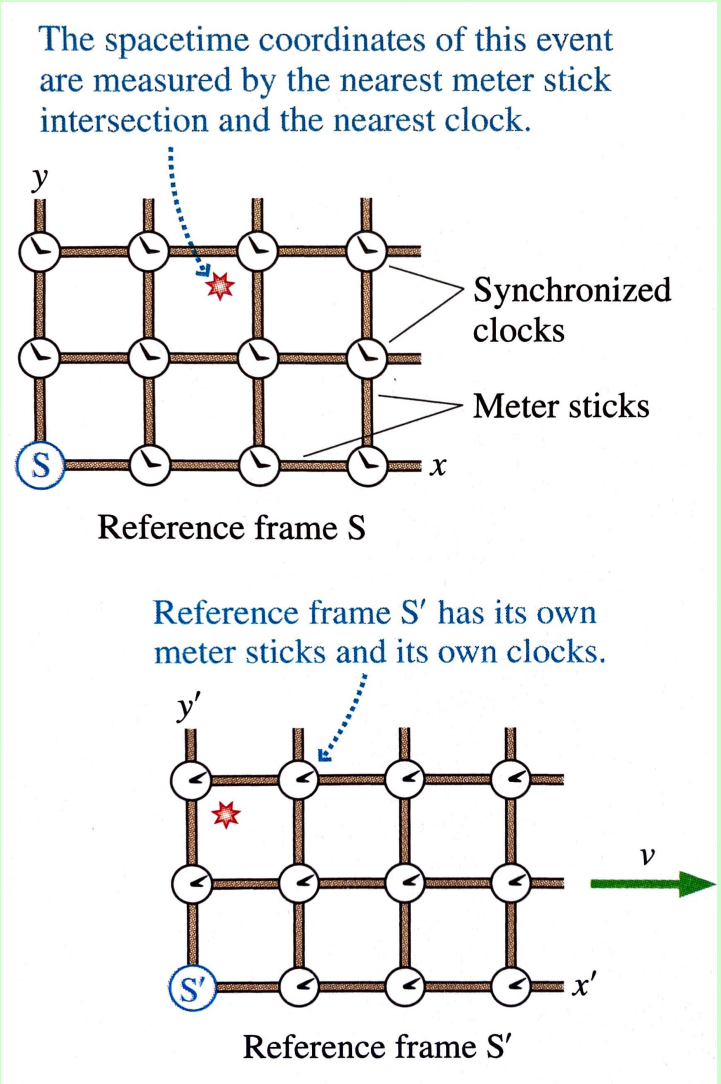


Light hits both ends at the same time



Light travels in both directions at c as the ship moves forward. Light hits back first.

Two observers won't necessarily agree on the order of events



Measuring events in spacetime

1. An Event is the (t,x,y,z) coordinate picked up by the nearest meterstick and clock.

2. Different reference frames have different values for each of these coordinates as measured by the nearest meterstick and clock in that frame

3. **Simultaneous Events** are when the synchronized clocks in a frame measure two events has happening at the same time. **Proper Length** is the distance between simultaneous events.

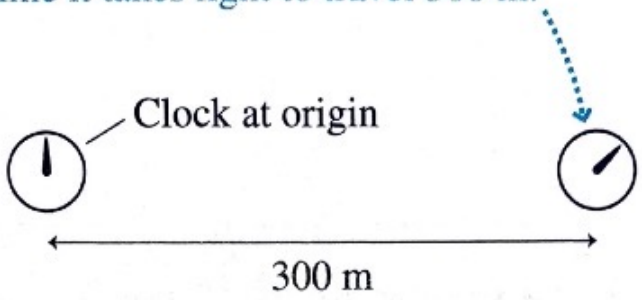
$$\boxed{\Delta t = t^{'} - t = 0} \quad \text{then} \quad \boxed{\Delta x = L_0}$$

4. **Proper Time** is the time interval measured by just one of these synched clocks. So:

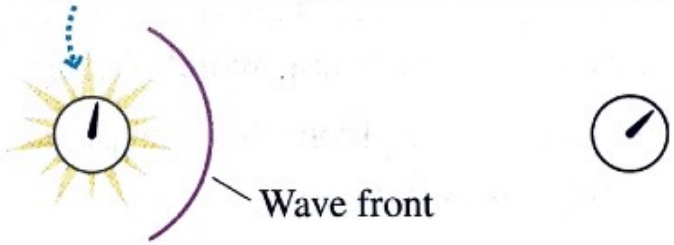
If $\boxed{\Delta x = 0}$ Then $\boxed{\Delta t = \Delta \tau}$

FIGURE 36.13 Synchronizing clocks.

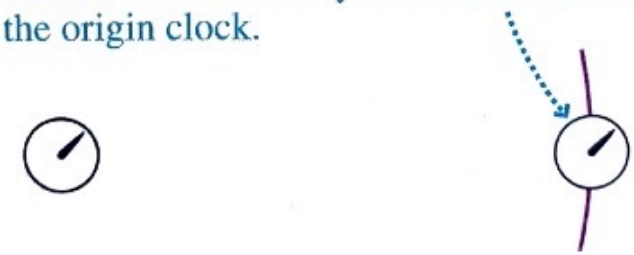
1. This clock is preset to 1.00 μs, the time it takes light to travel 300 m.



2. At $t = 0$ s, a light flashes at the origin and the origin clock starts running. A very short time later, seen here, a light wave has begun to move outward.



3. The clock starts when the light wave reaches it. It is now synchronized with the origin clock.



Ex. Andy is in S at the origin looking toward +x.
Betsy is at x =900 m looking toward –x
Firecracker explodes somewhere between them.
Andy records at t = 4 microseconds. Betsy records at t = 3 microseconds

What are the spacetime coordinates of the event?

Important note: Event isn't when an observer sees something that happens
an event is when and where it really does happen

Ex. Alberta is in S at the origin looking toward +x.
At t= 3 microseconds, she sees a firecracker 1 explode at x = 600 m
At t= 5 microseconds, she sees firecracker 2 explode at x = 1200 m.

A. Are the two events simultaneous according to Alberta?

B. Would all observers agree with the ordering of the two events?

C. What is the proper distance between the events?